EXP. NO: 08
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NMAP to Discover Live Host using Nmap Scans.

AIM:

To discover live hosts on a network using Nmap host discovery scans - ARP scan, ICMP Scan and TCP/UDP ping scan - and to understand which protocol operates at which layer of the TCP/IP model and when each one applies.

1. ARP host discovery

The ARP scan successfully identified the active hosts in the local subnet. ARP request were broadcast to the network, and the live host responded with their MAC address.

2. ICMP Echo Scan (-PE)

The ICMP echo scan identified host that responded to ICMP echo requests. The host that replied were considered Active.

3. ICMP timestamp scan (-PP)

The ICMP timestamp scan identified some live hosts that responded to timestamp requests some host's didn't responded because ICMP timestamp packets may be filter by firewalls.

4. ICMP address Mask scan (-PM)

The ICMP address mask scan received few or no responses. This is because address Mask request are commonly blocked or unsupported by modern systems and firewalls.

5. ICP SYN Ping (-PS) :-

The TCP SYN ping identified active hosts by sending TCP SYN packets to the specified port.

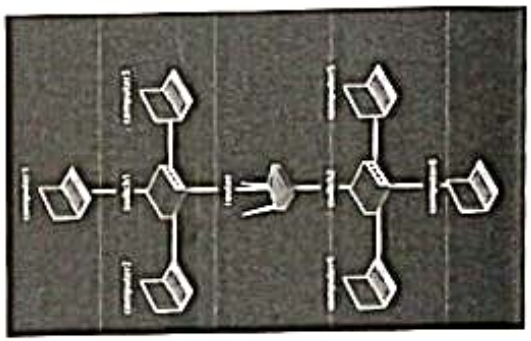
An SYN/ACK response from an open port or an RST response from a closed port or an RST response.

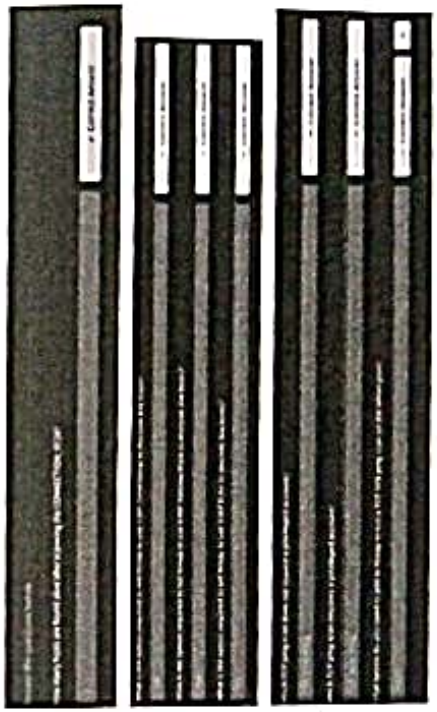
6. TCP ACK ping (-PA)

The TCP ACK ping identified active hosts by sending TCP packets with the ACK flag set. Send the packets does not belong to an existing connection a live host generally responds with an RST packets.

7. UDP ping

The UDP ping identified active host by sending UDP packets to the specified port. when the target port is closed, the host normally responds with an ICMP port unreachable message confirming that the host is active.





RESULT :

The NMAP host-discovery experiments were successfully performed using ARP, ICMP, TCP and UDP scanning techniques.